

MACHINE LEARNING FOR SIGNAL PROCESSING

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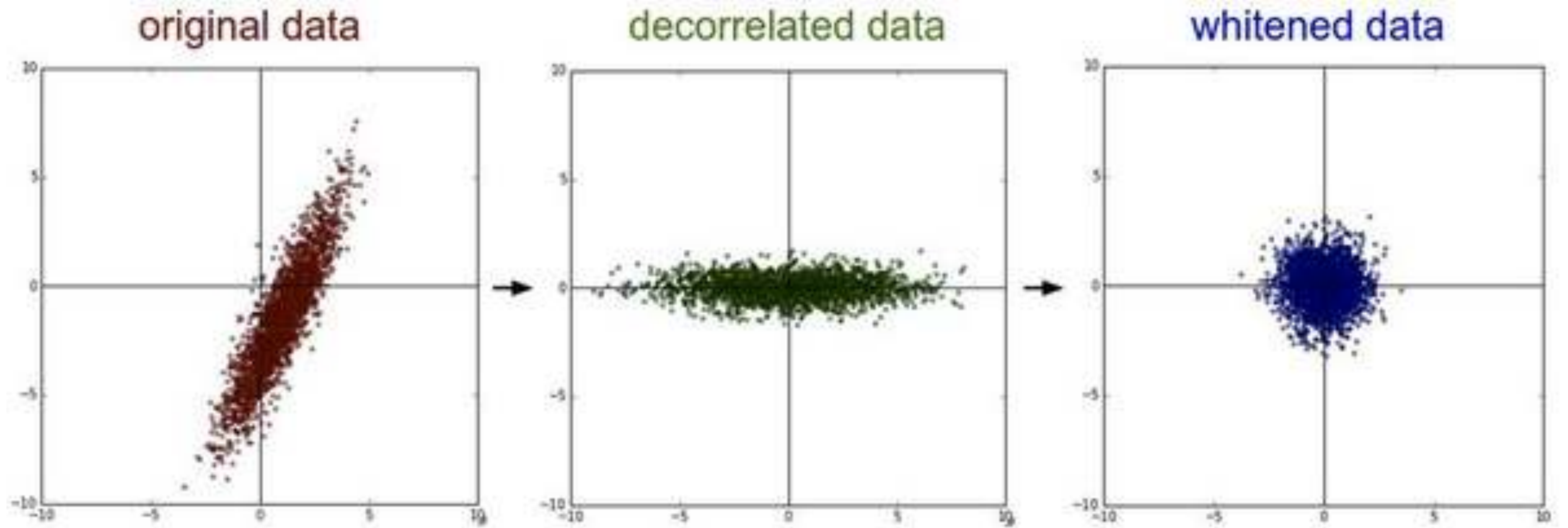
<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>



PRINCIPAL COMPONENT ANALYSIS

- ❖ Reducing the data $\mathbf{x}_n$ of dimension D to lower dimension
- ❖ Projecting the data into subspace which preserves maximum data variance
- ✓ Maximize variance in projected space $M < D$
- ❖ Equivalent formulated as minimizing the error between the original and projected data points.

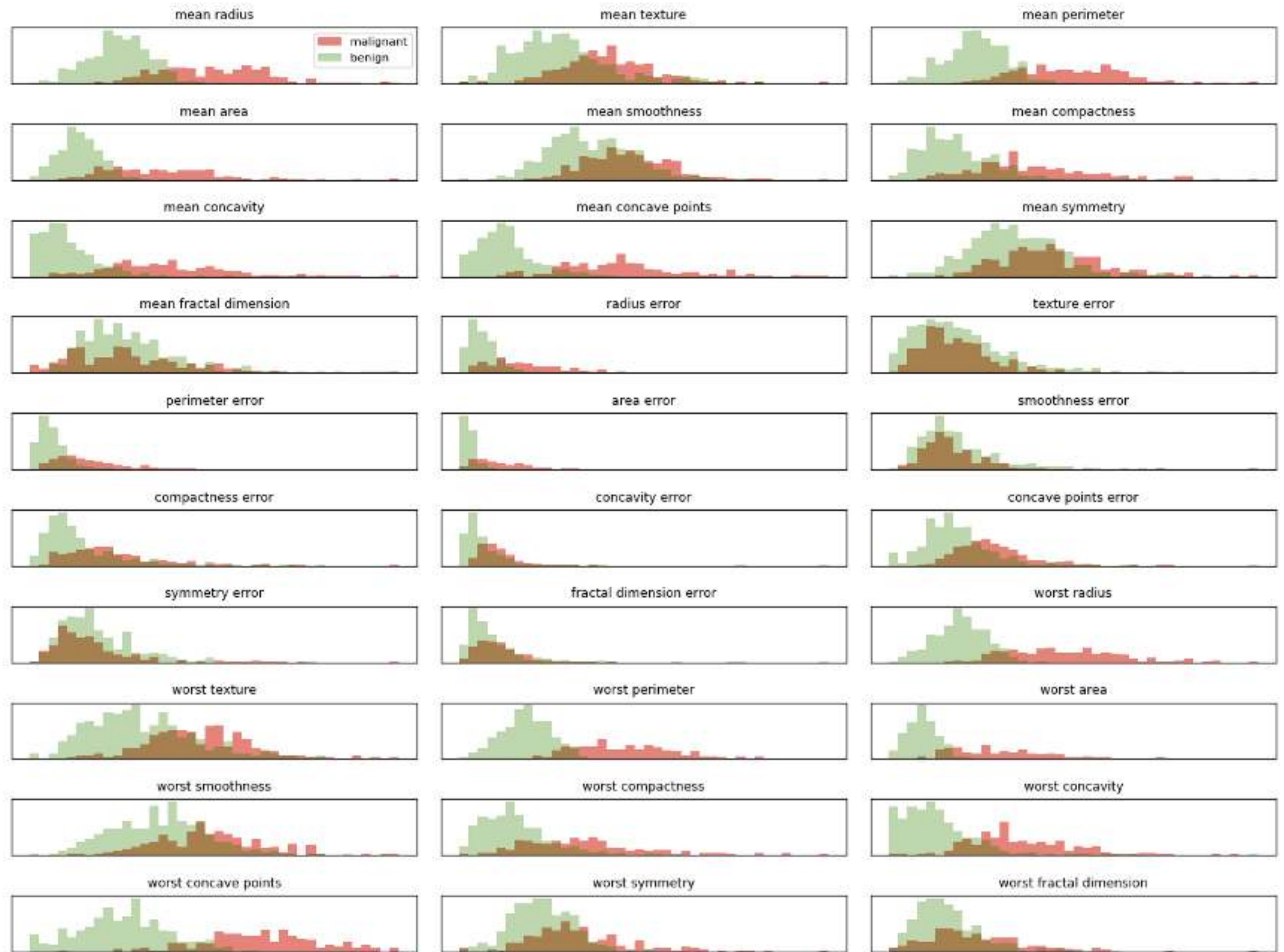
WHITENING VS DECORRELATIONS



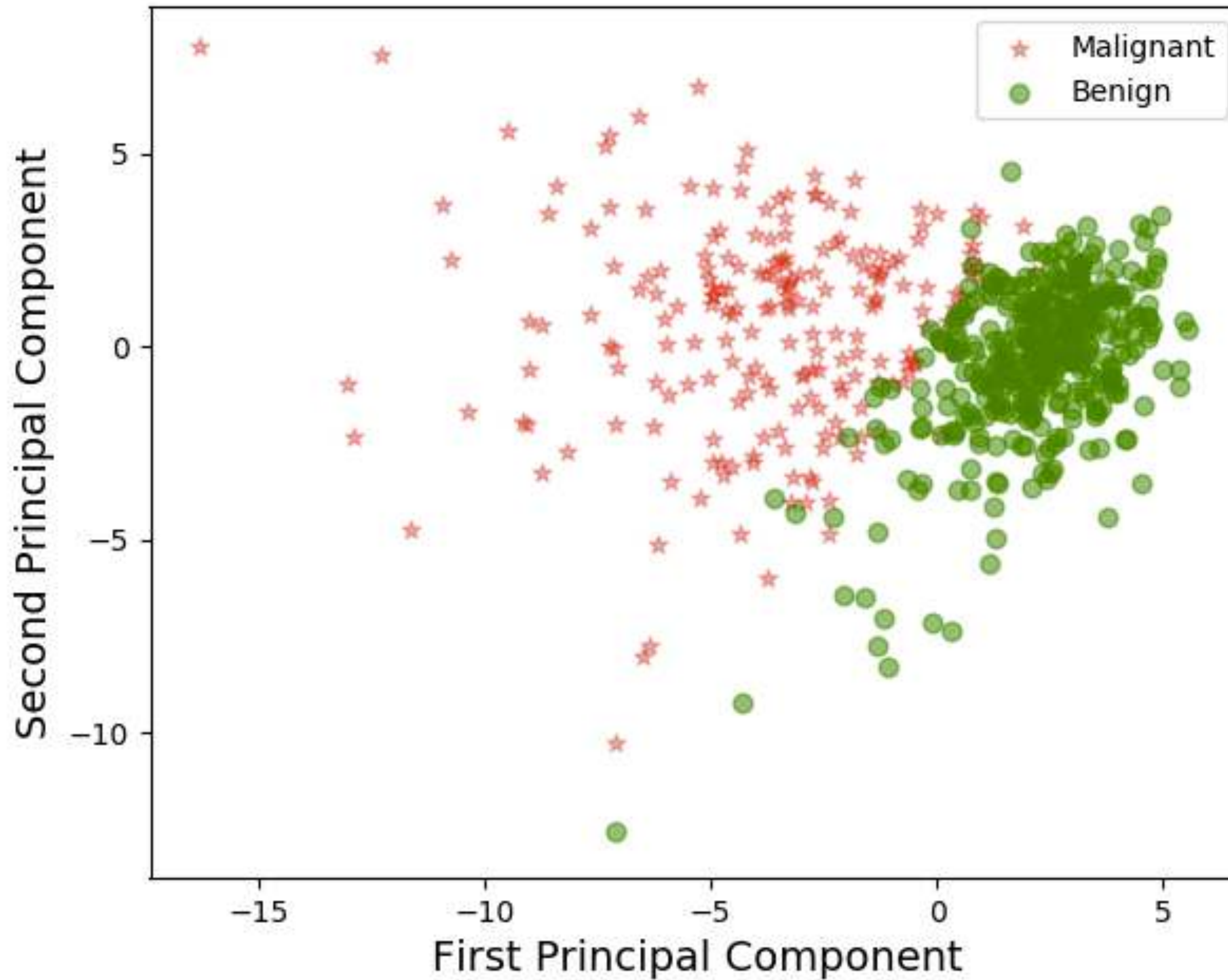
APPLICATION

- ❖ Wisconsin Cancer dataset ([https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+\(Diagnostic\)](https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)))
- ❖ 569 participants
- ❖ 212 (M) 357 (B)
- ❖ 30 features —> digitized image of a fine needle aspirate (FNA) of a breast mass. The features describe characteristics of the cell nuclei present in the image.

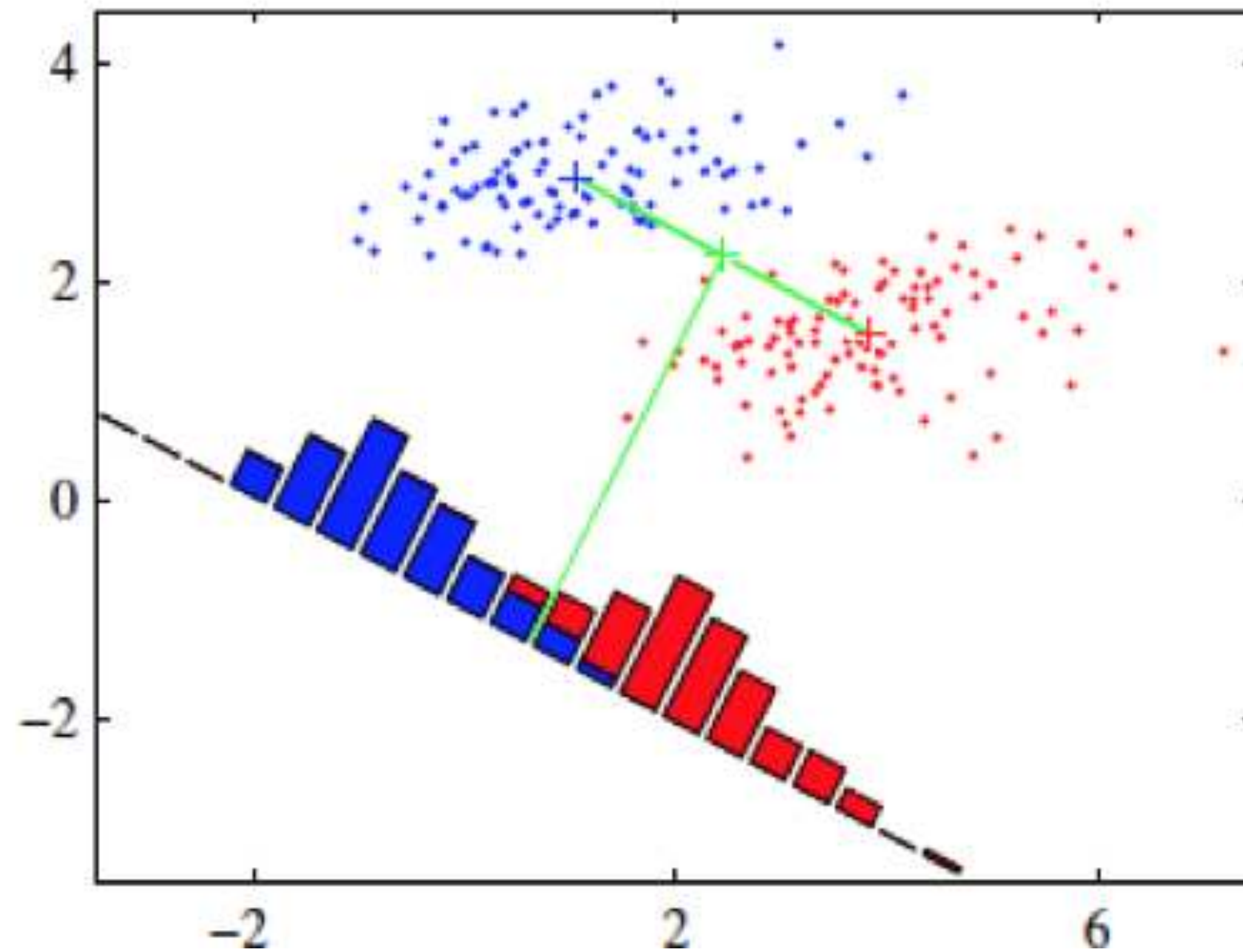
Raw Features



PCA



WITHOUT THE WITHIN CLASS FACTOR



LINEAR DISCRIMINANT ANALYSIS

❖ Generalized Eigenvalue problem

Find a linear transform $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$ with a criterion which maximizes the class separation

- Maximize the between class distance in the projected space while minimizing the within class covariance

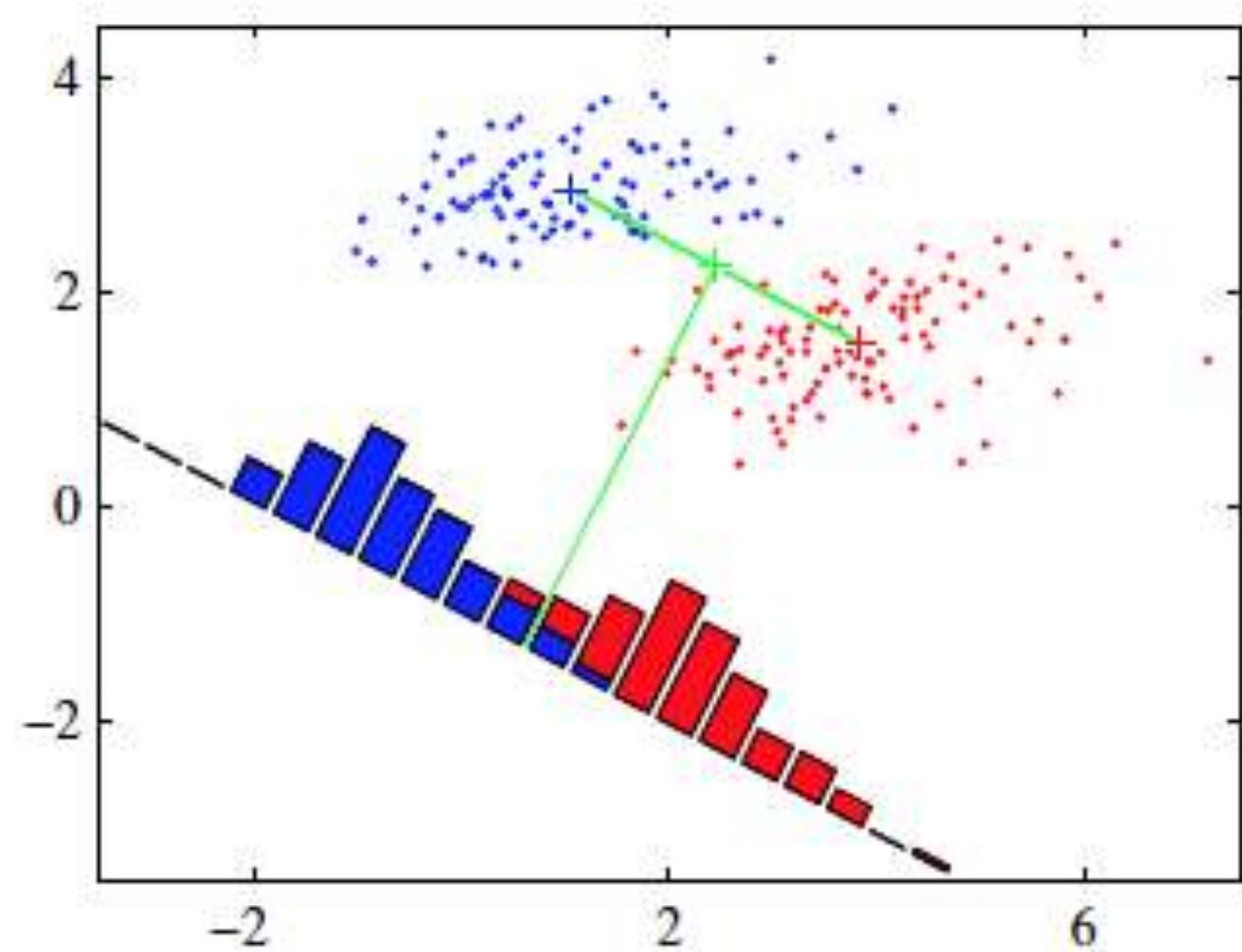
$$J = \frac{\mathbf{w}^T \mathbf{S}_b \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

$$\mathbf{S}_b = \sum_{k=1}^K N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T \quad \mathbf{S}_w = \sum_{k=1}^K \sum_{n \in C_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^T$$

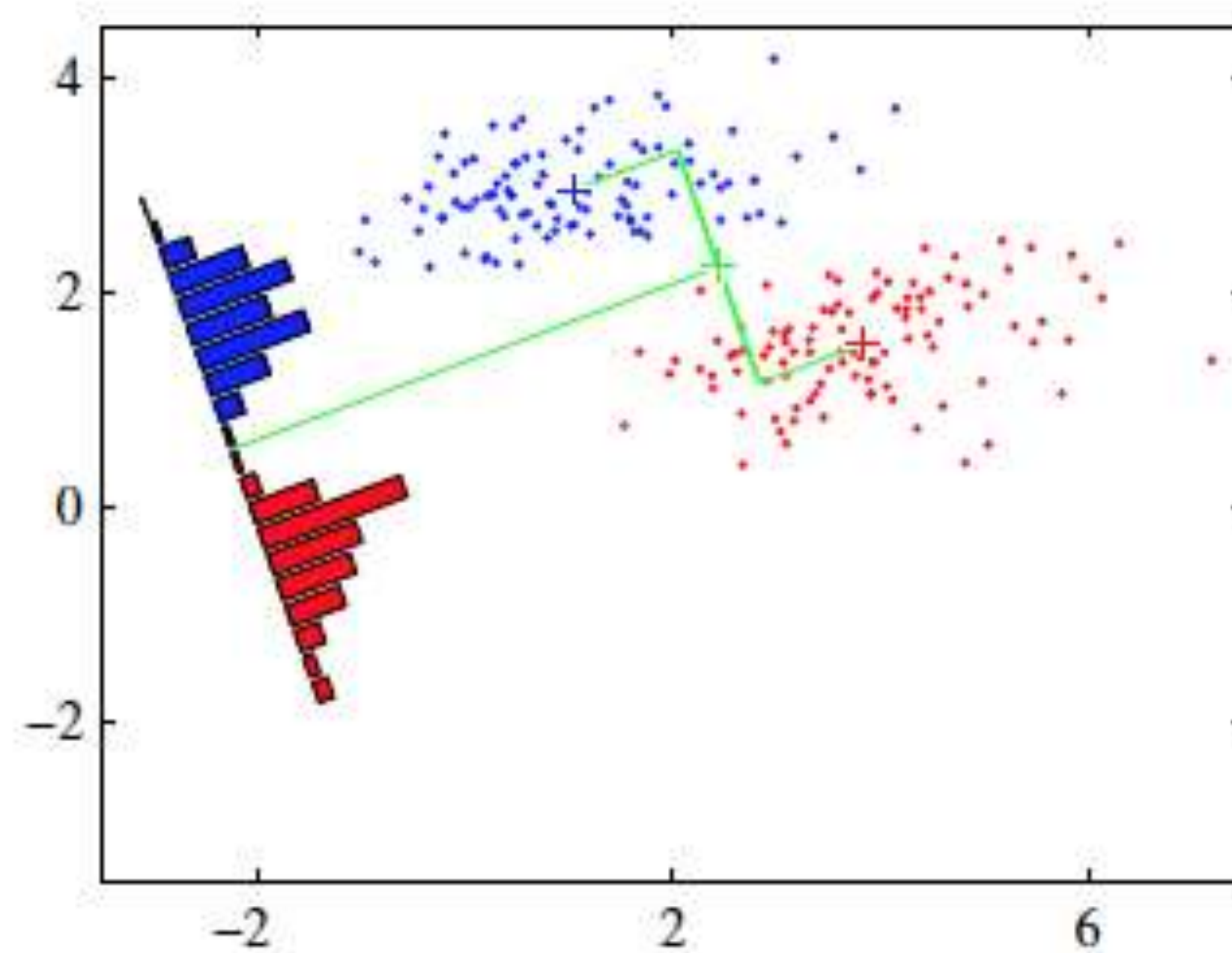
Eigen analysis of $\mathbf{S}_w^{-1} \mathbf{S}_b$

LINEAR DISCRIMINANT ANALYSIS

Projecting on line joining means



Fisher Discriminant

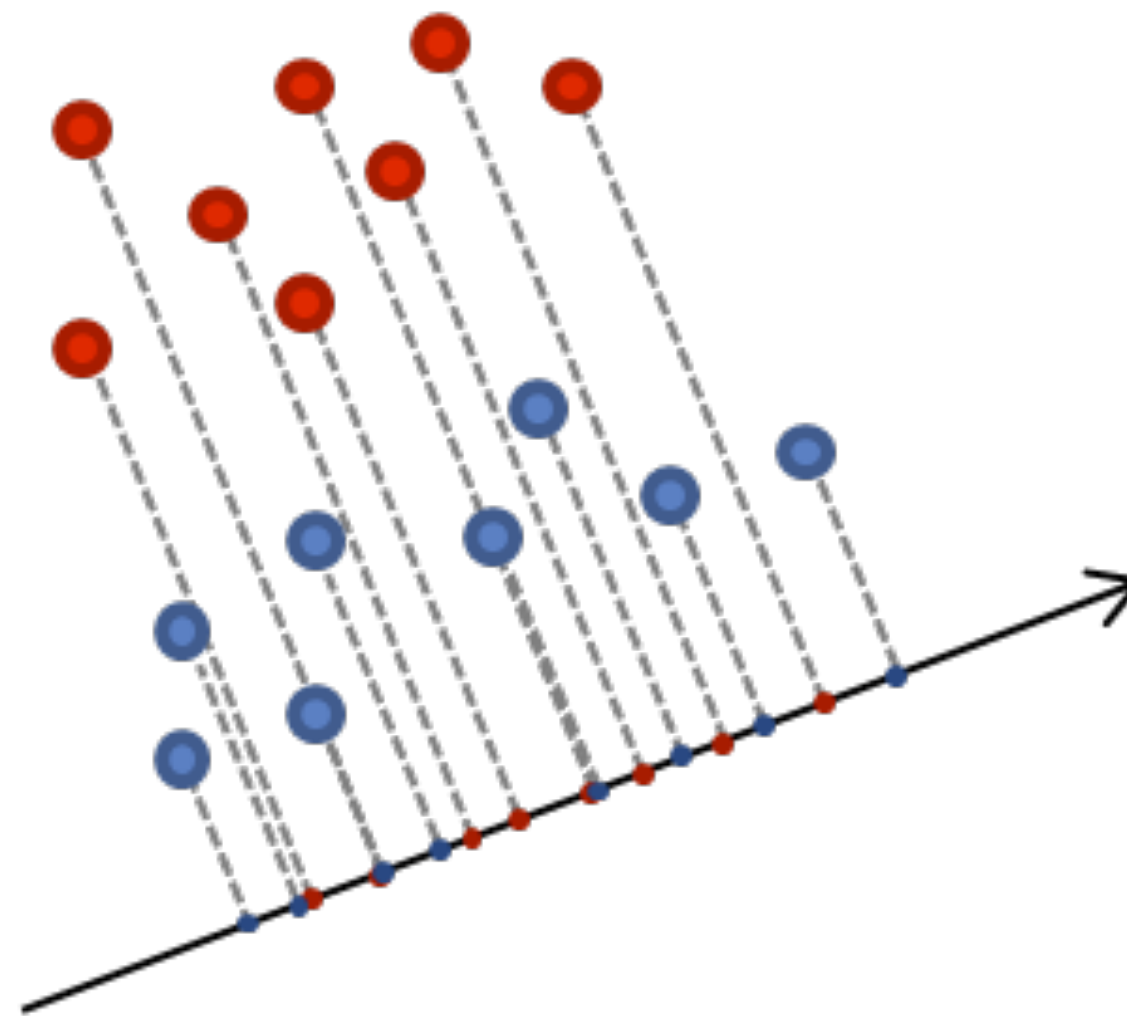


PCA VERSUS LDA

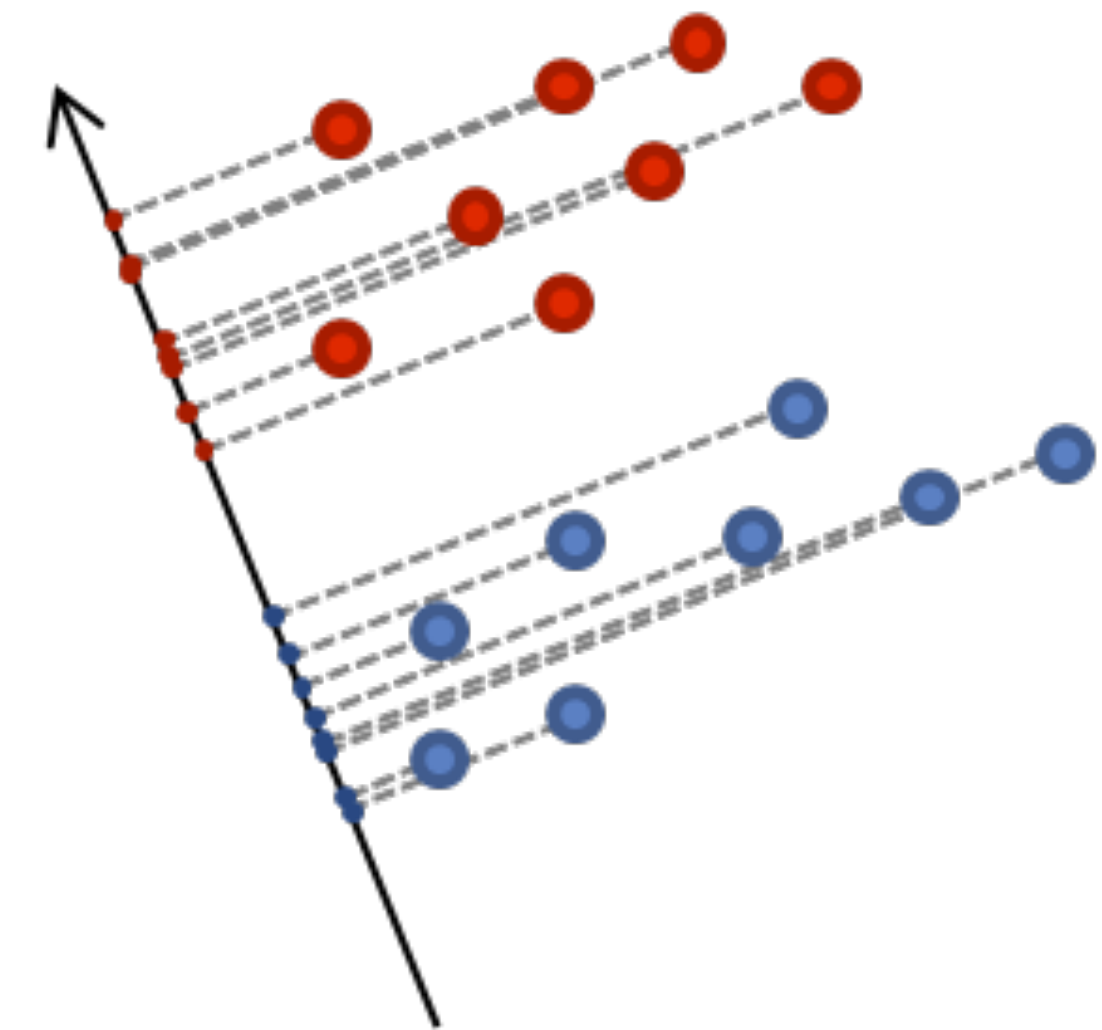
Labelled
data



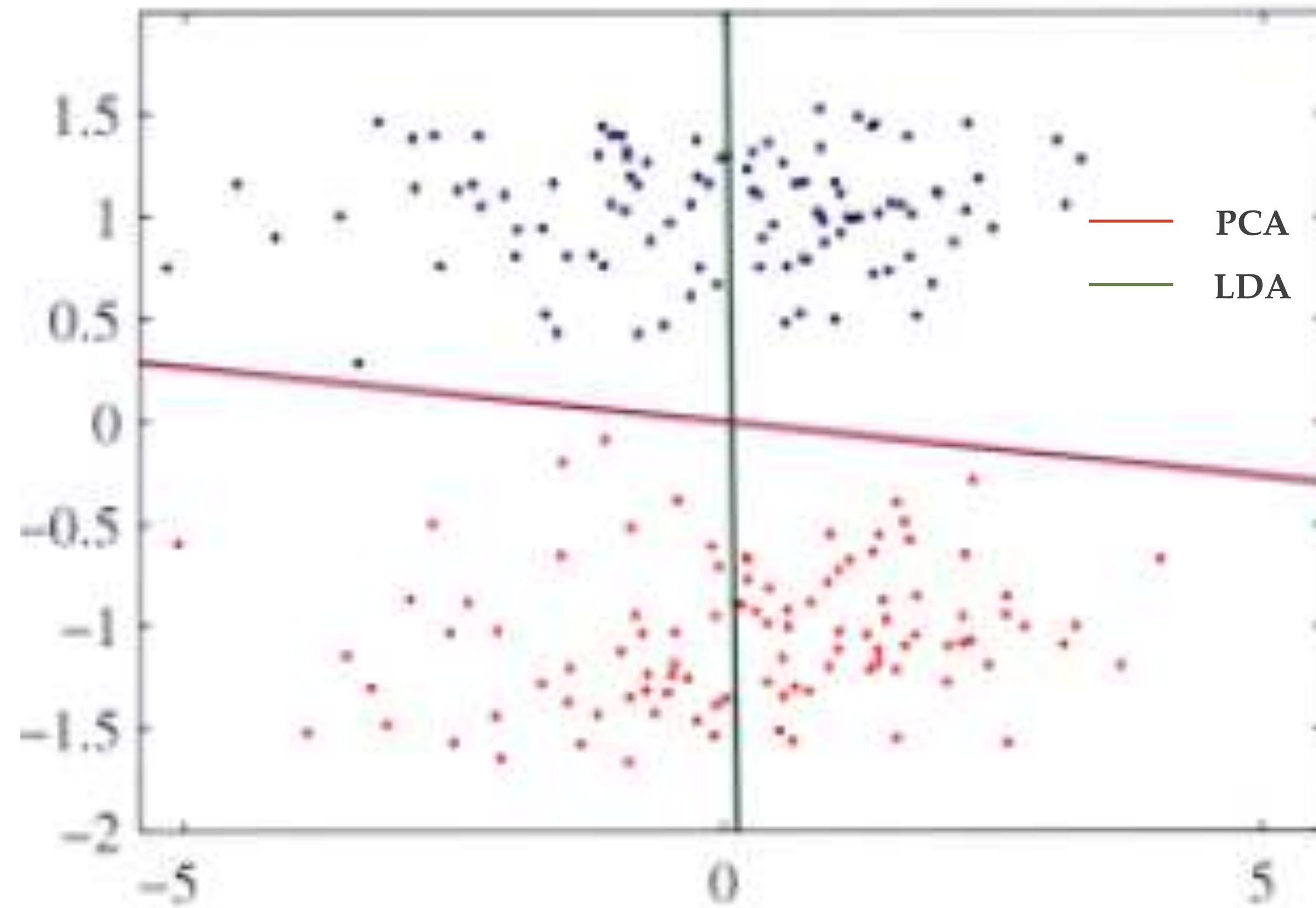
PCA projection:
Maximising the variance of
the whole set



LDA projection:
Maximising the distance
between groups



LINEAR DISCRIMINANT ANALYSIS



DECISION THEORY (PRML CHAP. 1.5)

❖ Decision Theory

✓ Inference problem

- ◉ Finding the joint density

✓ Decision problem

- ◉ Using the inference to make the classification or regression decision $p(\mathbf{x}, t)$

DECISION PROBLEM - CLASSIFICATION

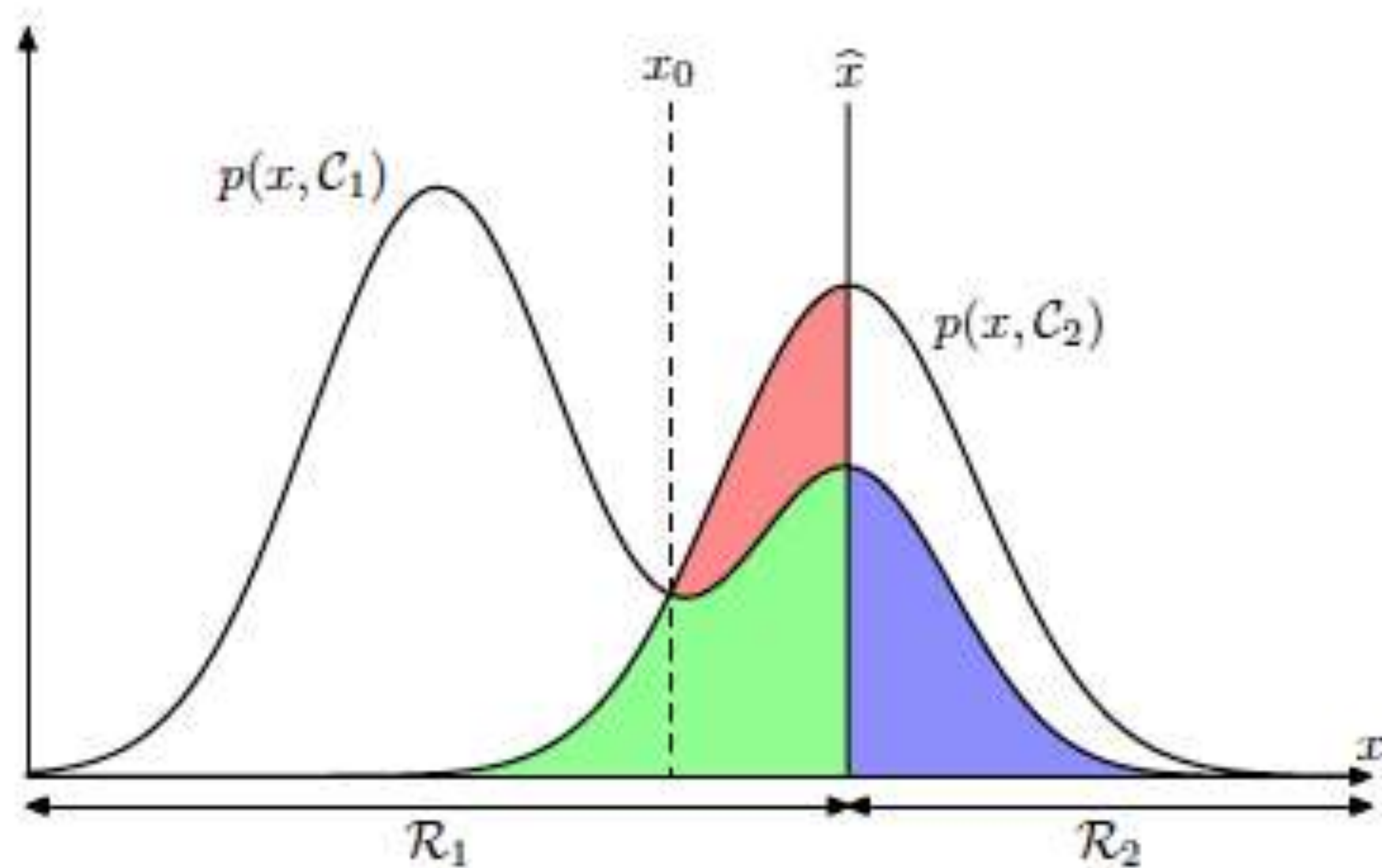
- ✓ Minimizing the mis-classification error
- ✓ Decision based on maximum posteriors

$$\operatorname{argmax}_j p(C_j|\mathbf{x})$$

- ✓ Loss matrix
 - ◉ Minimizing the expected loss

$$\operatorname{argmax}_j \sum_k L_{k,j} p(C_k|\mathbf{x})$$

VISUALIZING THE MAX. POSTERIOR CLASSIFIER



APPROACHES FOR INFERENCE AND DECISION

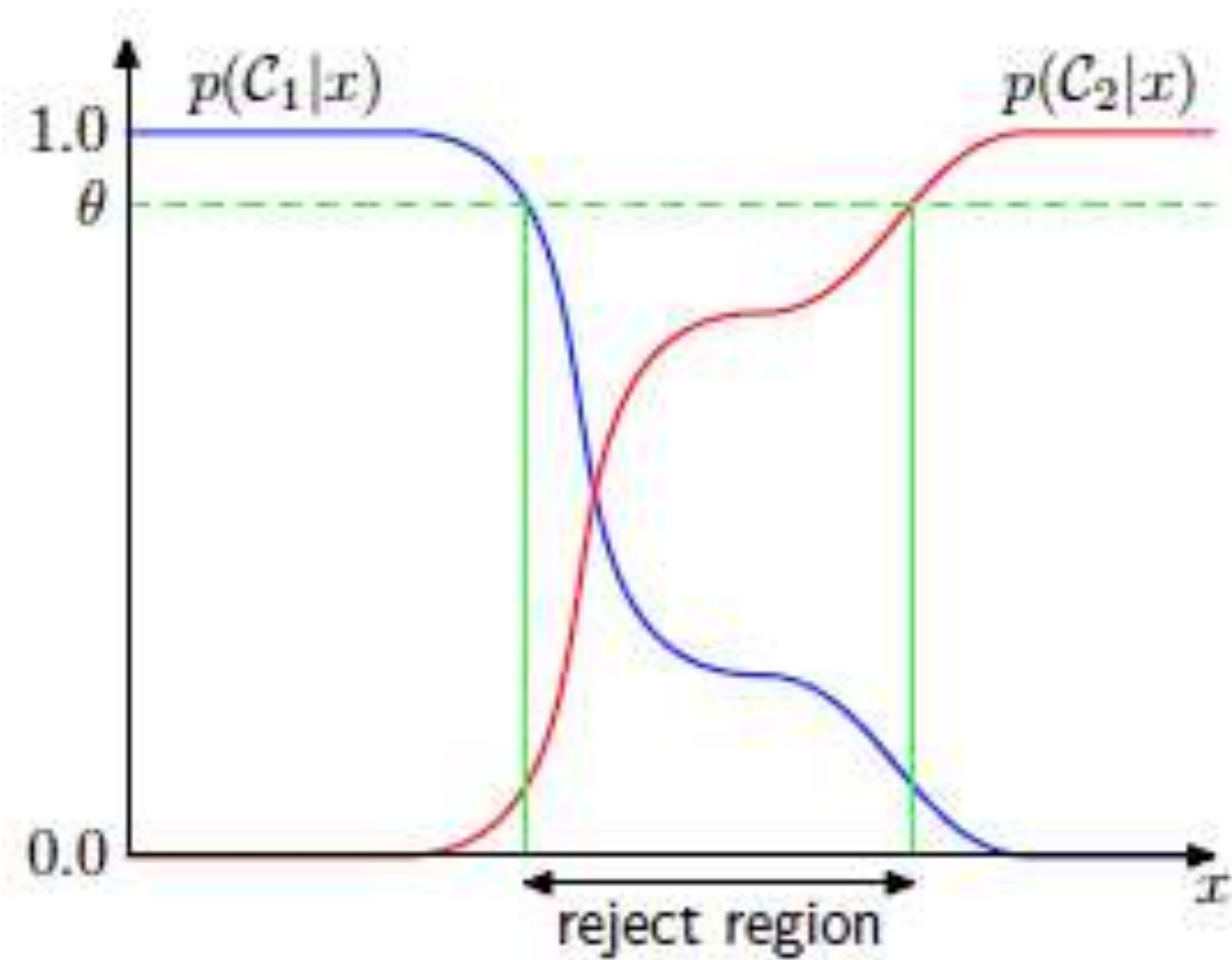
I. Finding the joint density from the data.

II. Finding the posteriors directly.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

III. Using discriminant functions for classification.

ADVANTAGE OF POSTERIORIORS

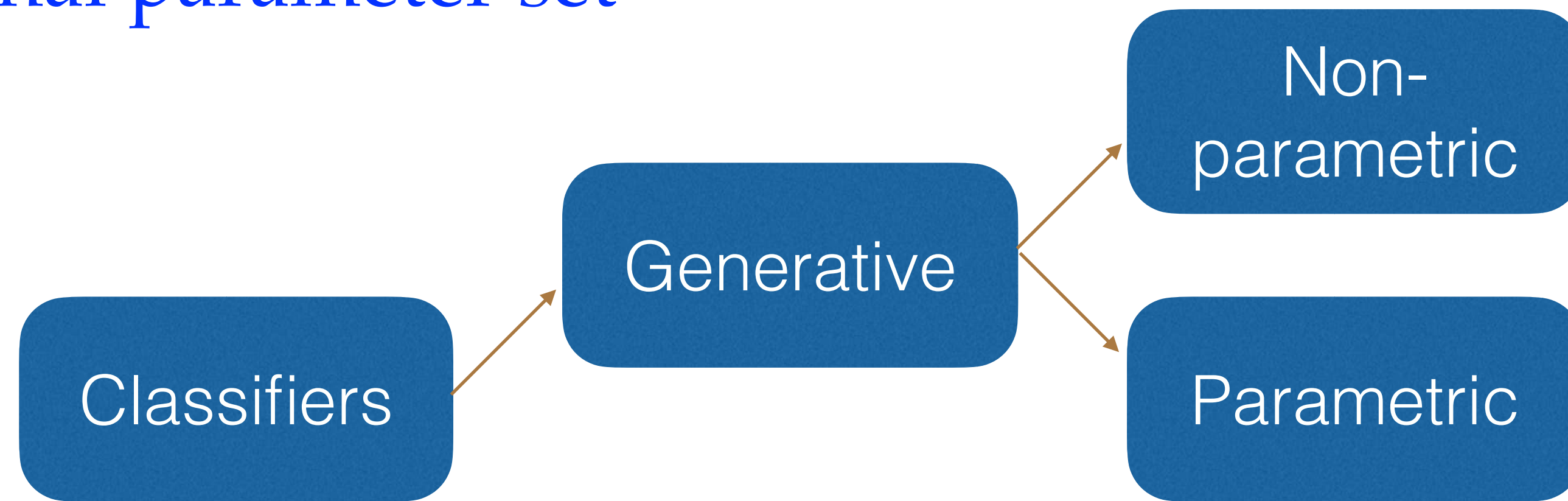


Decision Rule for Regression

- ❖ Minimum mean square error loss
- ❖ Solution is conditional expectation.

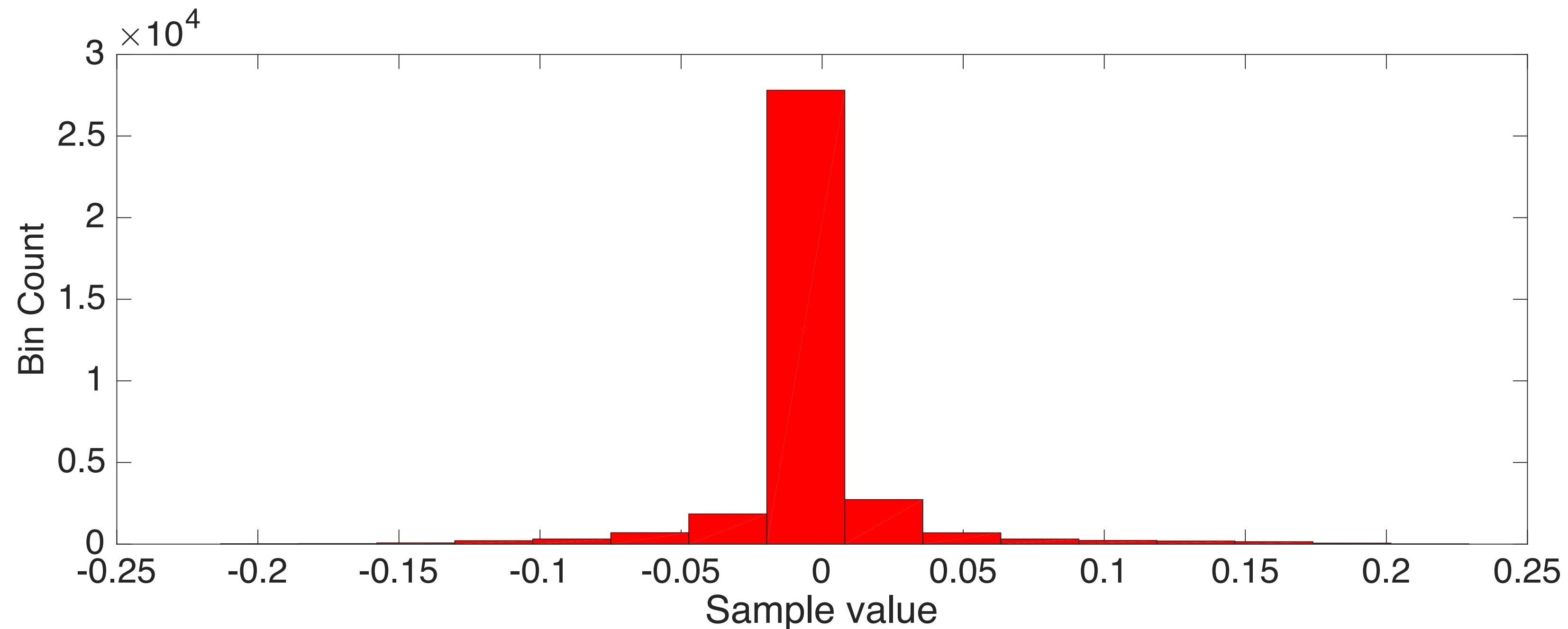
GENERATIVE MODELING

- ❖ Collection of probability distributions which are described by a **finite dimensional parameter set**



Non-parametric Modeling

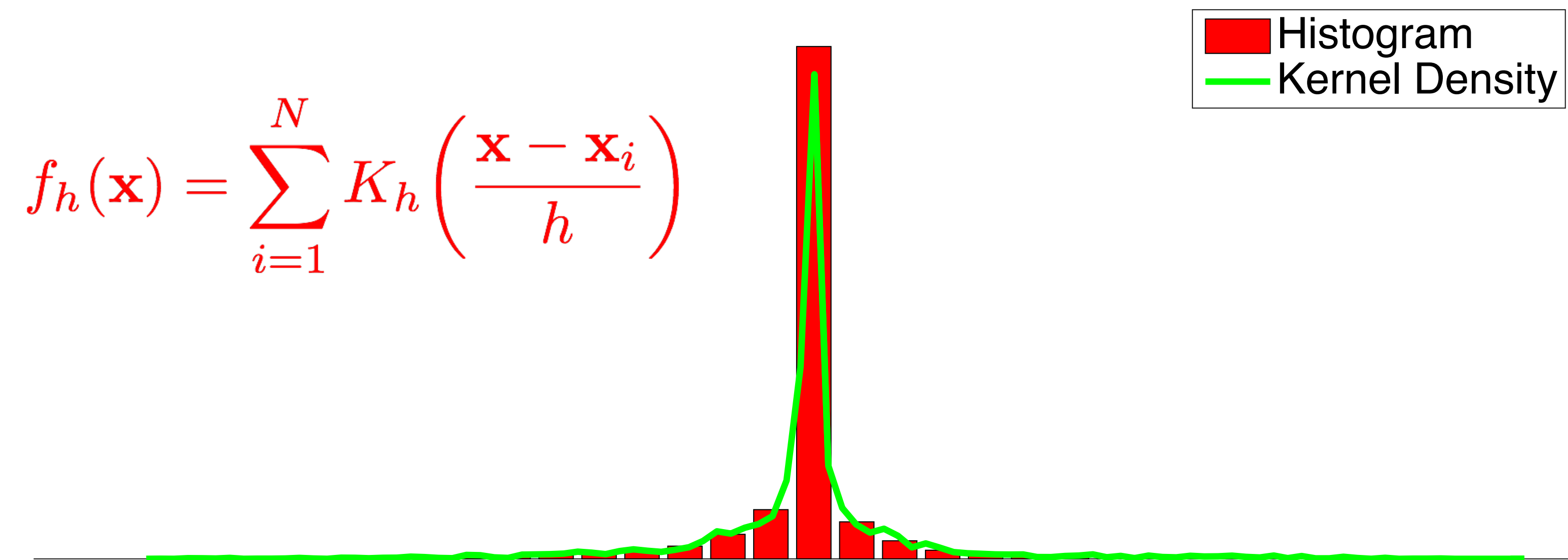
- **Non-parametric** models do not specify an apriori set of parameters to model the distribution. Example - Histogram



The density is not smooth and has block like shape.

Non-parametric Modeling

- **Non-parametric** models do not specify an apriori set of parameters to model the distribution.
- Example - Kernel Density Estimators



Kernel is a smooth function which obeys certain properties

Non-parametric Modeling

- Non-parametric methods are dependent on number of data points
 - Estimation is difficult for large datasets.
- Likelihood computation and model comparisons are hard.
- Limited use in classifiers

Parametric Models (Chap 2 PRML)

- ❖ Collection of probability distributions which are described by a **finite dimensional parameter set**

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K) \quad P = \{P_{\boldsymbol{\theta}}\}$$

- Examples -

- Poisson Distribution

$$p_{\lambda}(j) = \frac{\lambda^j}{j!} e^{-\lambda}$$

- Bernoulli Distribution

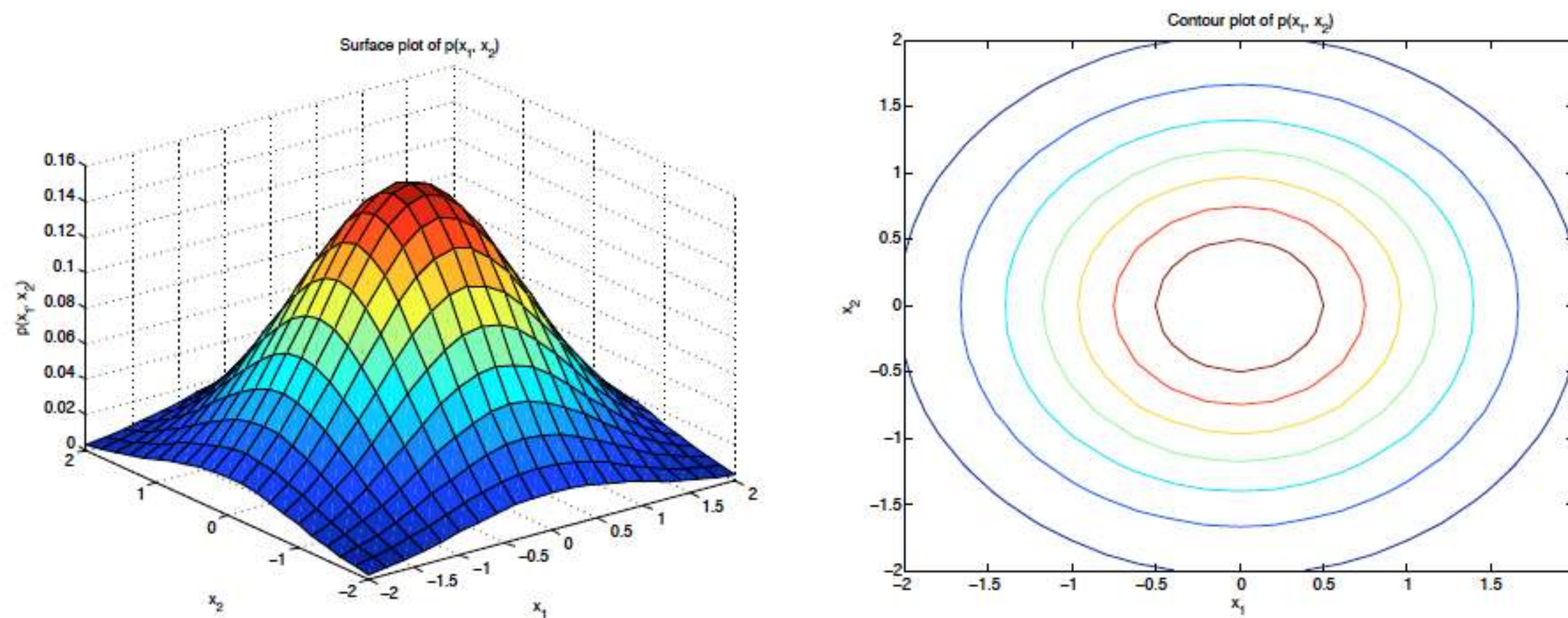
$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{i=1}^D \mu_i^{x_i} (1 - \mu_i)^{1-x_i}$$

- Gaussian Distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Gaussian Distribution

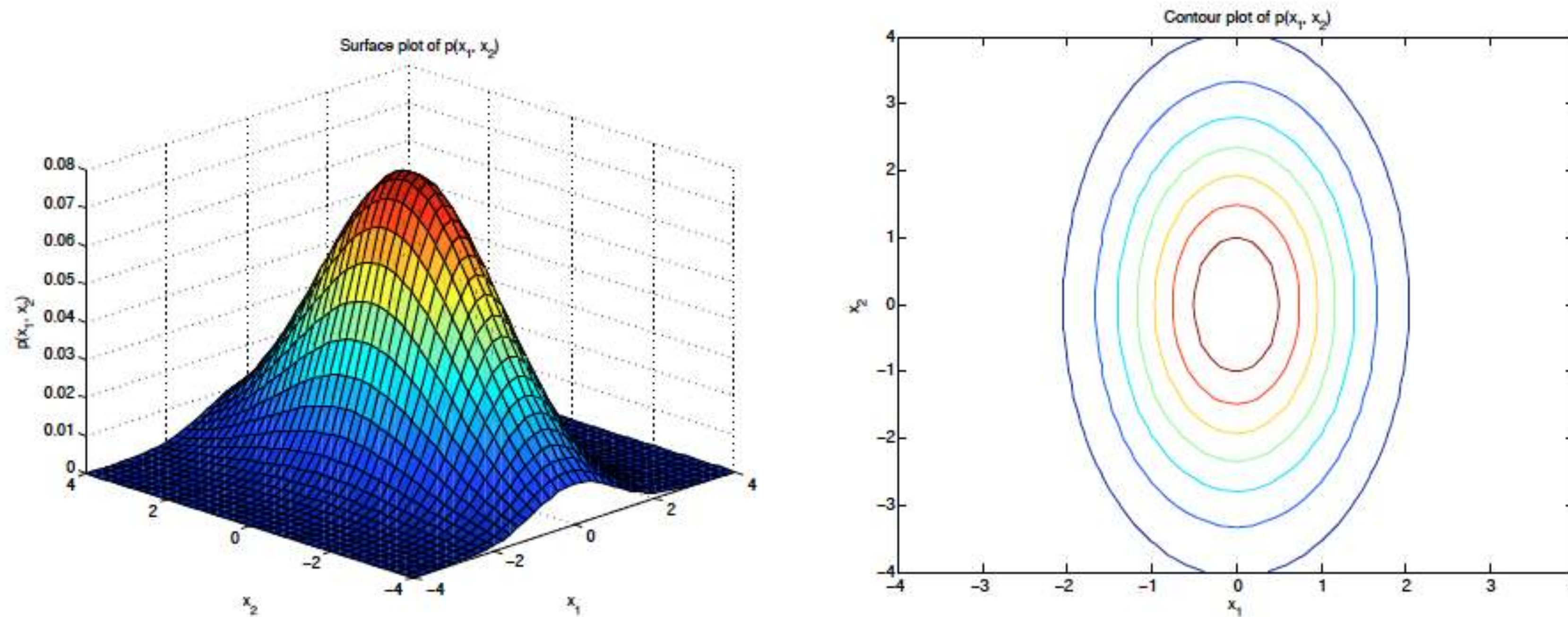
One of most widely used and well studied model



Points of equal probability lie on on contour
Diagonal Gaussian with Identical Variance

Gaussian Distribution

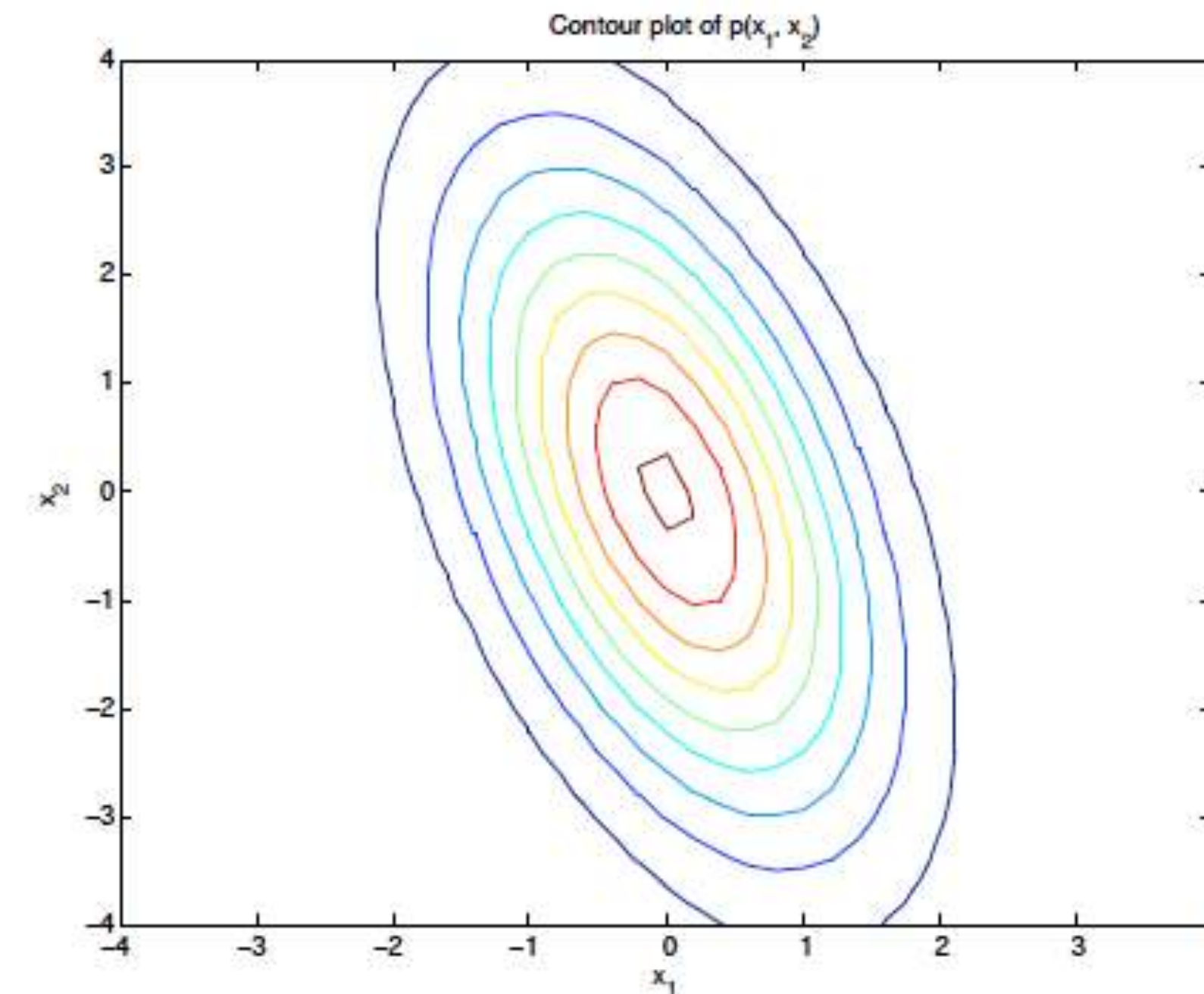
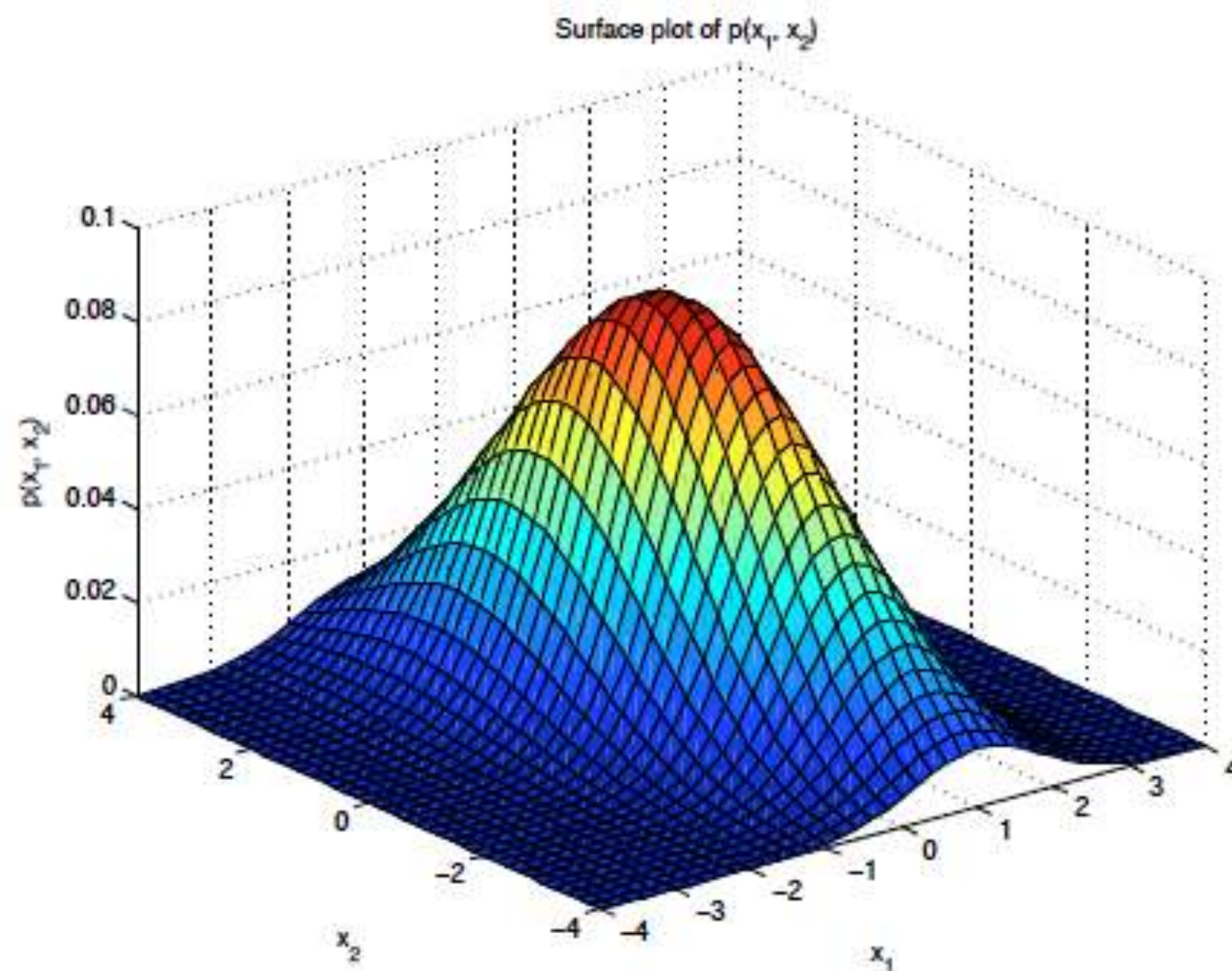
Insights into two dimensional Gaussian distribution



Diagonal Gaussian with different variance

Gaussian Distribution

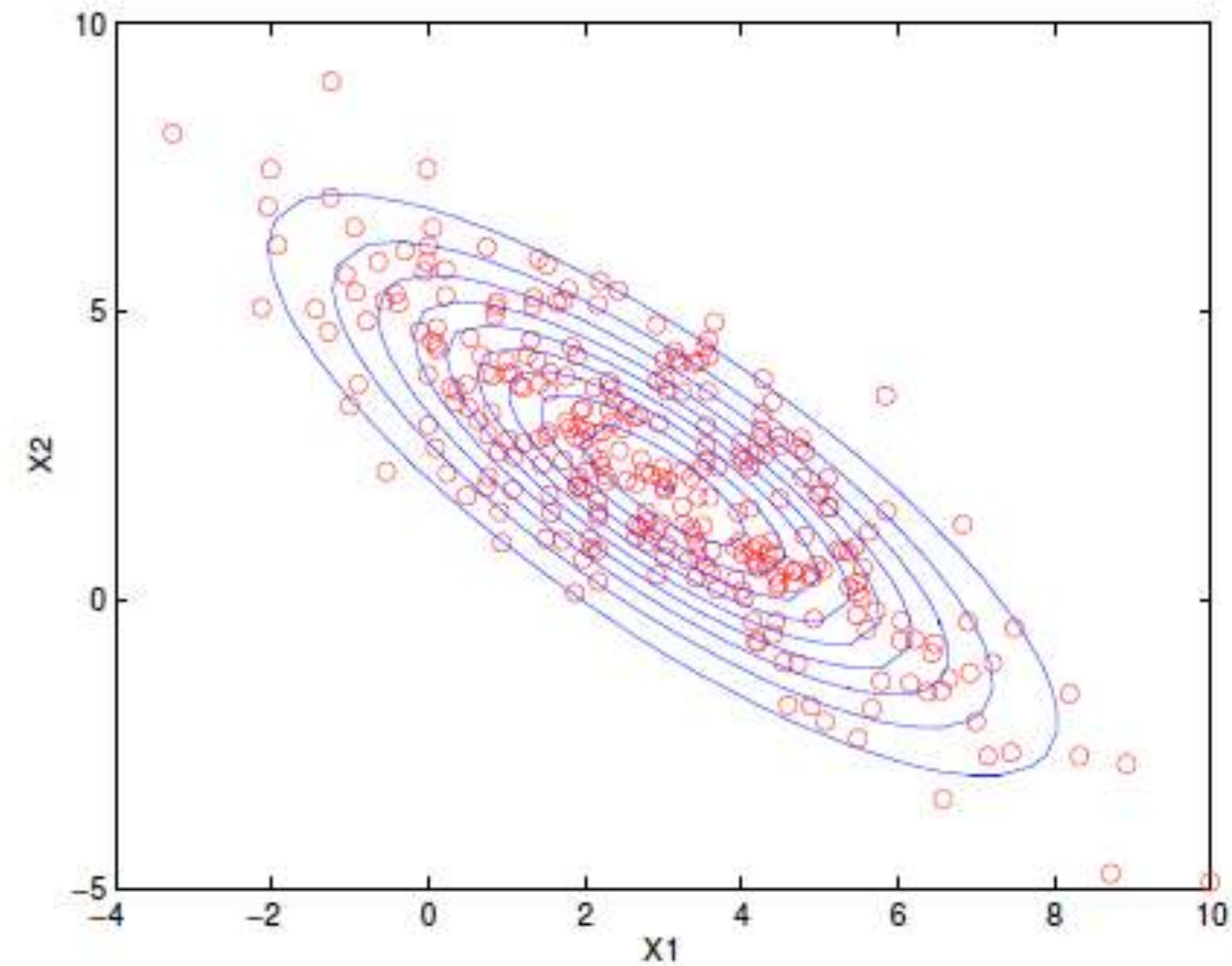
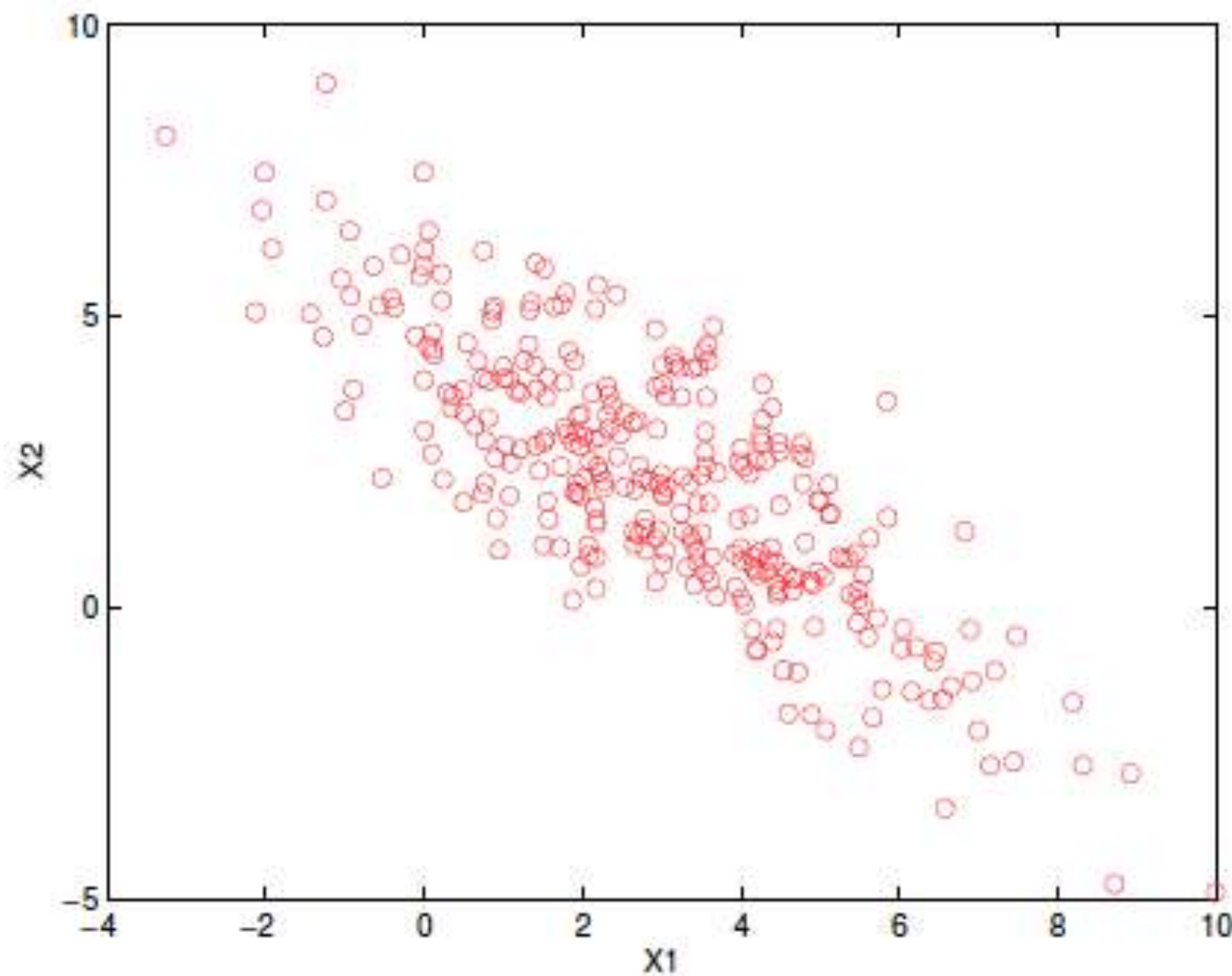
Insights into two dimensional Gaussian Distribution



Full covariance Gaussian distribution

Gaussian Distribution

Fitting the data with a Gaussian Model



Finding the parameters of the Model

- ✓ The Gaussian model has the following parameters

$$\theta = (\mu, \Sigma)$$

- ✓ Total number of parameters to be learned for D dimensional data is $D^2 + D$
- ✓ Given N data points $\{\mathbf{x}_i\}_{i=1}^N$ how do we estimate the parameters of model.
 - ➔ Several criteria can be used
 - ➔ The most popular method is the maximum likelihood estimation (MLE).

MLE

Define the likelihood function as $L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i | \boldsymbol{\theta})$

The maximum likelihood estimator (MLE) is

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{arg\,max}} L(\boldsymbol{\theta})$$

The MLE satisfies nice properties like

- Consistency (convergence to true value)
- Efficiency (has the least Mean squared error).

MLE

For the Gaussian distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp\left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$$L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i|\boldsymbol{\theta})$$

$$\log L(\boldsymbol{\theta}) = -\frac{ND}{2} - \frac{N}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{i=1}^N \left((\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right)$$

To estimate the parameters $\frac{\partial \log L}{\partial \boldsymbol{\mu}} = 0$

THANK YOU

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